Supplementary Information

Bayesian Neural Surrogates Unify Uncertainty Propagation,
Sensitivity Attribution and Calibration for Coupled
Multi-Physics Reactor Analysis

补充信息

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1 Supplementary Figures / 补充图

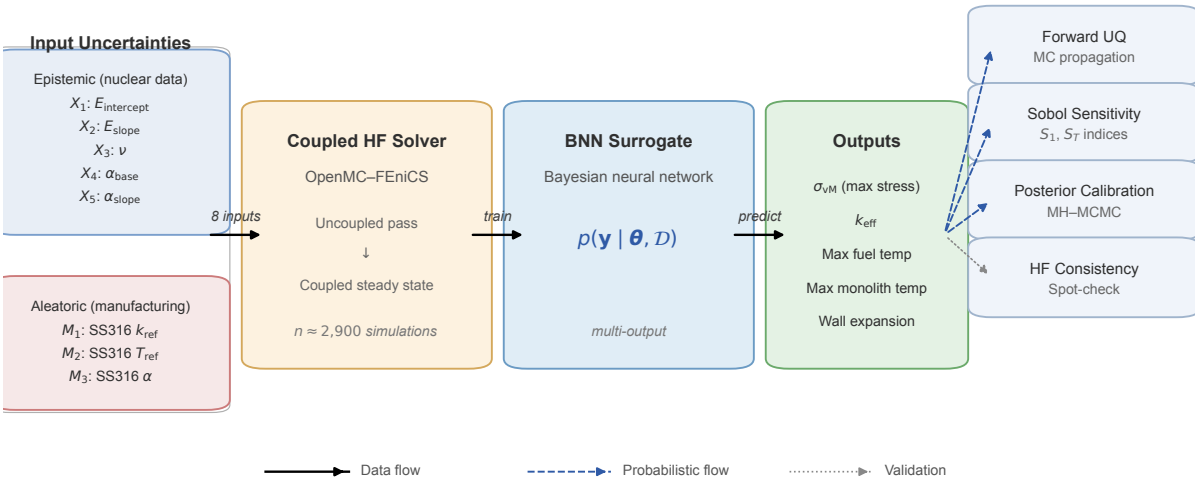

Figure 1: **Fig. S1 | BNN surrogate workflow.** Schematic of the end-to-end probabilistic analysis framework. The BNN surrogate is trained on coupled OpenMC–FEniCS data and produces a posterior predictive distribution. This single distribution serves as the shared probabilistic layer for forward uncertainty propagation, Sobol sensitivity attribution, and MCMC posterior calibration, without retraining between analyses.
图 S1 | BNN 代理工作流示意图。端到端概率分析框架。BNN 代理在耦合 OpenMC–FEniCS 数据上训练，产生后验预测分布，作为三类分析的共享概率层。

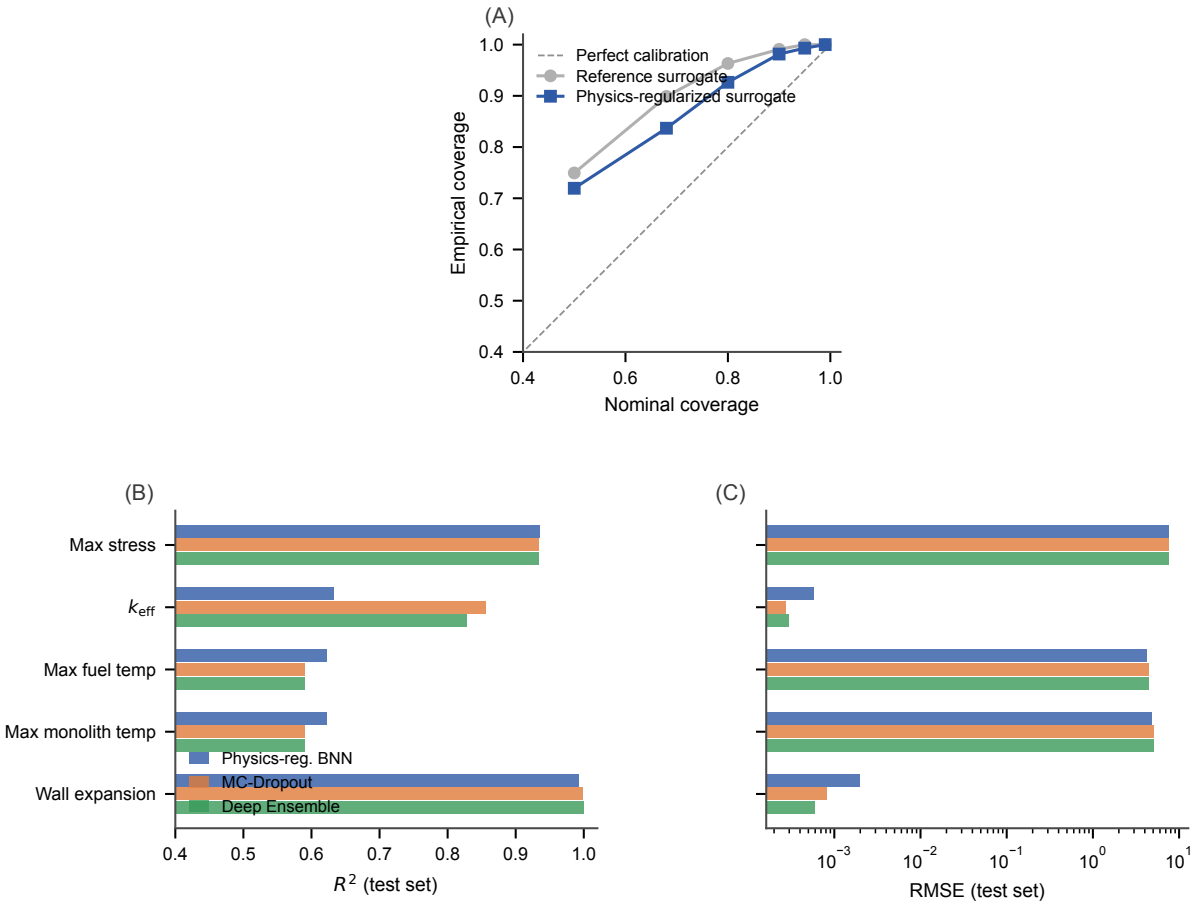

Figure 2: **Fig. S2 | Model validation: parity and residual analysis.** Predicted versus true values and residual distributions for primary outputs across all four BNN variants (435 test samples). Shaded regions show the 90% predictive interval.
图 S2 | 模型验证：一致性与残差分析。四种 BNN 变体在 435 个测试样本上的预测值与真实值及残差分布。

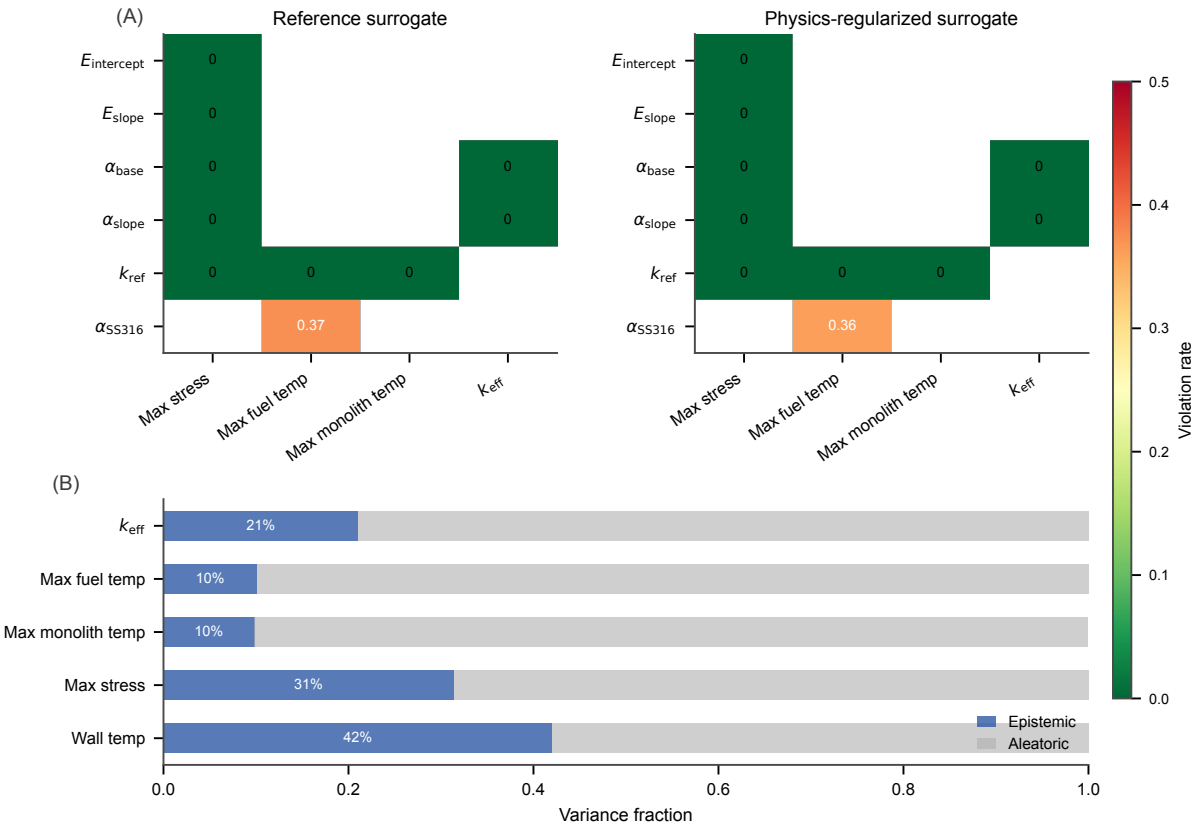

Figure 3: **Fig. S3 | Physics consistency and uncertainty decomposition.** (**A**) Monotonicity violation rates for input–output pairs across four BNN variants. (**B**) Epistemic vs. aleatoric variance decomposition. Aleatoric uncertainty dominates across all outputs (58–90% of total variance).
图 S3 | 物理一致性与不确定性分解。(A) 四种 BNN 变体的单调性违反率。(B) 认知与偶然方差分解。

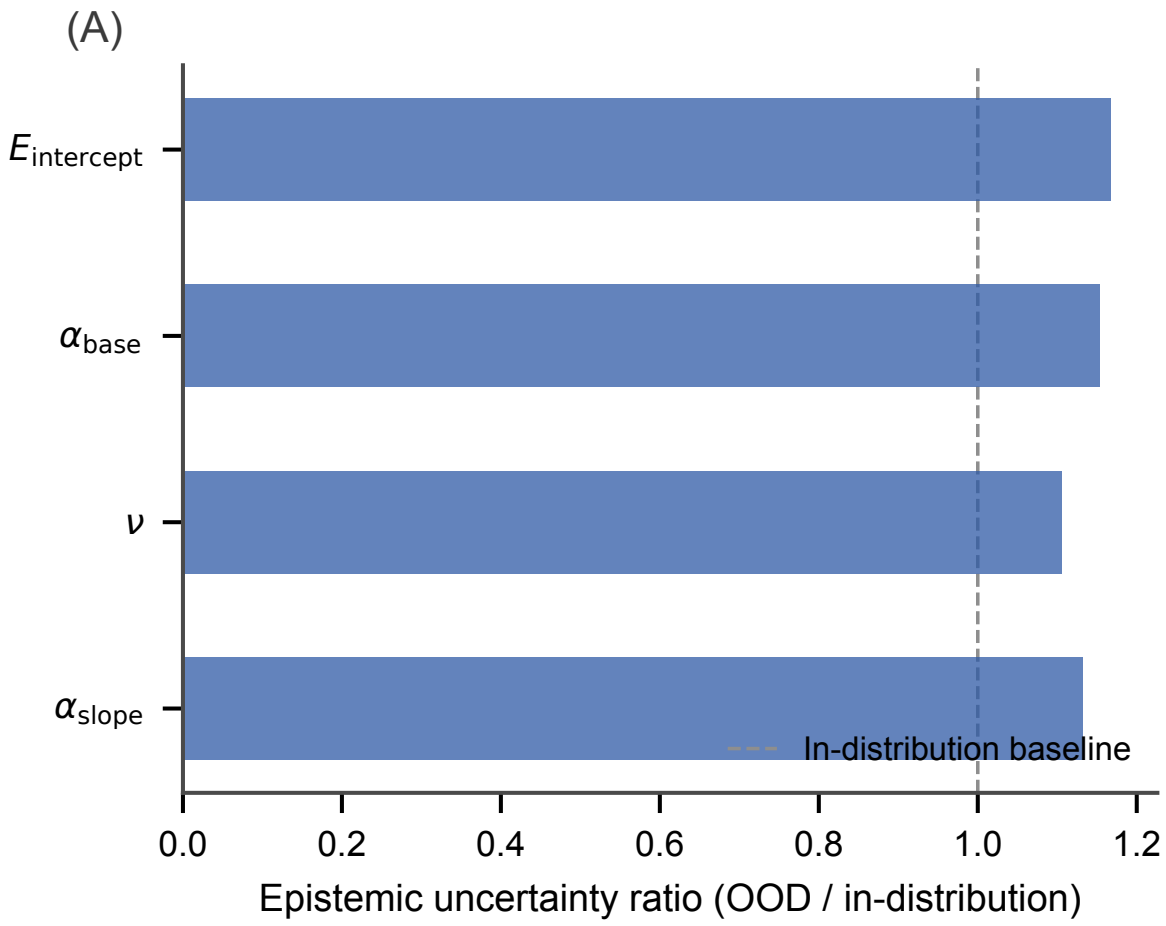

Figure 4: **Fig. S4 | Out-of-distribution (OOD) coverage and epistemic uncertainty.** 90% predictive interval coverage for in-distribution versus four feature-tail subsets. Stress retains coverage above 0.75 across all tails. k_{eff} coverage degrades substantially on the α_{base} tail (R^2 drops to approximately 0.29), indicating that the Sobol and posterior conclusions for k_{eff} are valid only within the training range of this dominant parameter. Elevated epistemic ratios on the α_{base} tail confirm that the BNN correctly signals increased model uncertainty outside the training domain.
图 S4 | 分布外覆盖率与认知不确定性。分布内与四个特征尾部子集的覆盖率对比。k_{eff} 在 α_{base} 尾部显著下降。

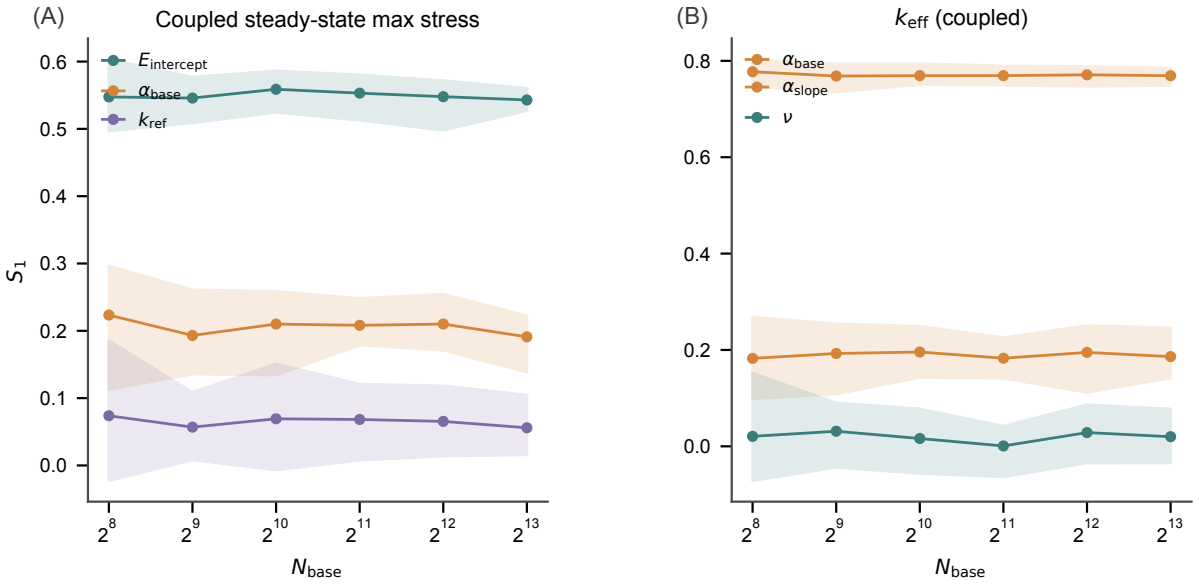

Figure 5: **Fig. S5 | Sobol index convergence with sample size.** First-order Sobol index S_1 as a function of sample size N_S (256 to 4096) for stress and k_{eff}, with 90% CI bands across 50 replications. Indices stabilise by $N_S = 2048$, confirming convergence at the chosen $N_S = 4096$.
图 S5 | Sobol 指标随样本量的收敛。应力和 k_{eff} 的一阶 Sobol 指标 S_1 随 N_S 变化，含 50 次重复的 90% CI 带。

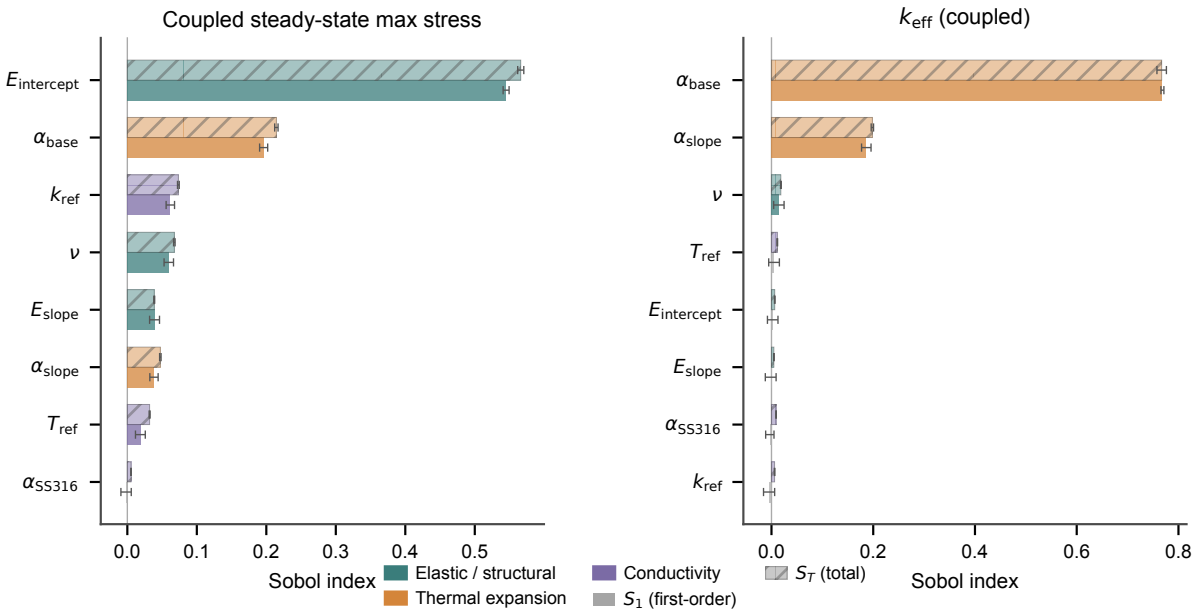

Figure 6: **Fig. S6 | Detailed Sobol sensitivity analysis.** First-order and total-order Sobol indices for all eight material parameters across all five primary outputs, showing the dominance of $E_{\text{intercept}}$ for stress and α_{base} for k_{eff}.
图 S6 | 详细 Sobol 敏感性分析。全部八个材料参数在五个主要输出上的一阶和全阶 Sobol 指标。

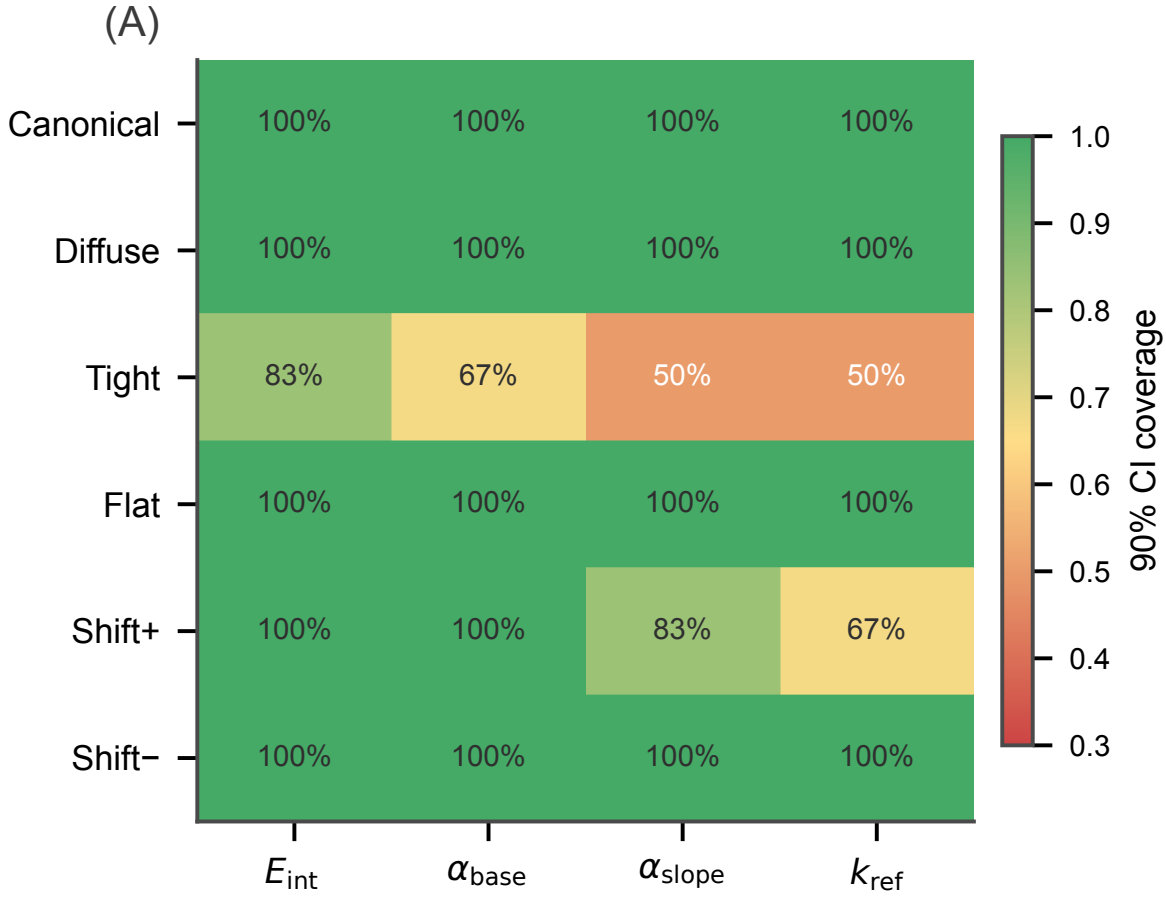

Figure 7: **Fig. S7 | Prior sensitivity analysis.** Posterior marginal distributions for the four calibrated parameters under six prior configurations: canonical, diffuse (2×), tight (0.5×), flat (uniform), shift$_+$ ($\mu + 0.5\sigma$), shift$_-$ ($\mu - 0.5\sigma$). The tight prior over-constrains the posterior, reducing coverage to 50–83% (Table S11).
图 S7 | 先验敏感性分析。六种先验配置下四个标定参数的后验边缘分布。紧凑先验过度约束后验。

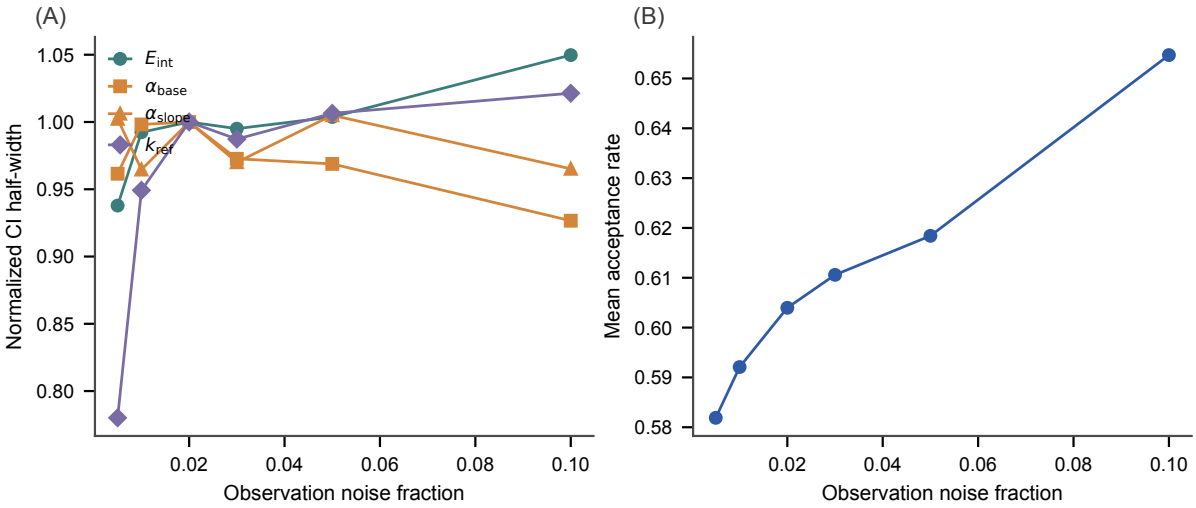

Figure 8: **Fig. S8 | Observation noise sensitivity.** Posterior width and acceptance rate as a function of assumed observation noise fraction (0.5%–10%). Posterior standard deviations increase monotonically; acceptance rates increase appropriately as the likelihood broadens (Table S12).
图 S8 | 观测噪声敏感性。后验宽度和接受率随假设观测噪声分数（0.5%–10%）的变化。

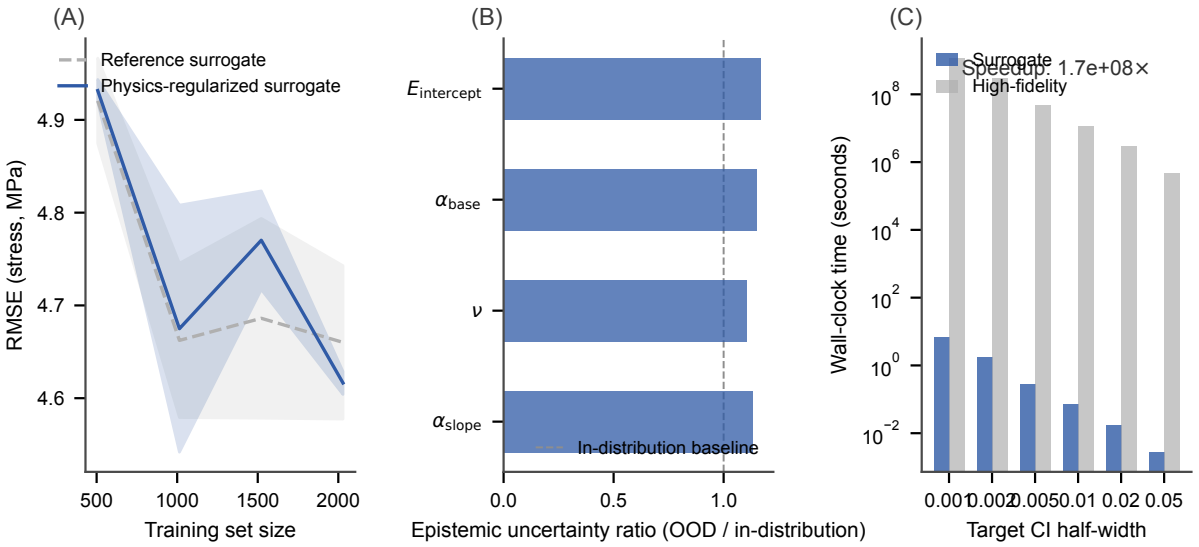

Figure 9: **Fig. S9 | Data efficiency and computational cost.** Test RMSE and R^2 as a function of training set size for the physics-regularized BNN. Near-saturated performance is achieved at 25% data utilisation (507 training samples; Table S13).
图 S9 | 数据效率与计算成本。物理正则化 BNN 的测试 RMSE 和 R^2 随训练集大小的变化。25% 数据即达到近饱和性能。

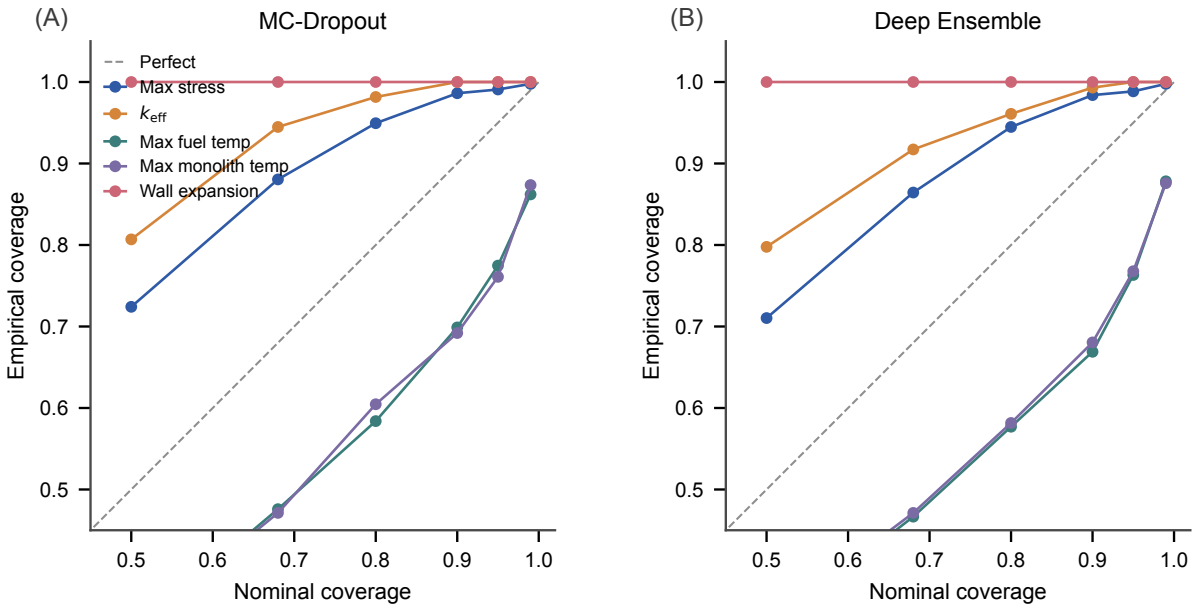

Figure 10: **Fig. S10 | External baseline calibration comparison.** Calibration curves comparing BNN, MC-Dropout and deep ensemble across primary outputs. The BNN achieves substantially better probabilistic calibration (ECE 0.027 vs. 0.18–0.19) on temperature outputs.
图 S10 | 外部基线校准对比。BNN、MC-Dropout 和深度集成的校准曲线。BNN 在温度输出上校准质量显著更优。

2 Supplementary Tables / 补充表

Table 1: **Table S1 | Material parameters: definitions, nominal values and priors.**
表 S1 | 材料参数：定义、标称值和先验。

Parameter	Symbol	Nominal value	Unit	Prior range ($\pm 10\%$)
Young's mod. intercept	$E_{\text{intercept}}$	2×10^{11}	Pa	$[1.8, 2.2] \times 10^{11}$
Young's mod. slope	E_{slope}	-7×10^{7}	Pa/K	$[-7.7, -6.3] \times 10^{7}$
Poisson's ratio	ν	0.31	—	[0.279, 0.341]
CTE base	α_{base}	1×10^{-5}	K^{-1}	$[0.9, 1.1] \times 10^{-5}$
CTE slope	α_{slope}	5×10^{-9}	K^{-2}	$[4.5, 5.5] \times 10^{-9}$
Ref. conductivity	k_{ref}	23.2	W/(m·K)	[20.88, 25.52]
Ref. temperature	T_{ref}	923.15	K	[830.84, 1015.47]
Cond. slope	k_{slope}	1/75	W/(m·K^2)	[0.012, 0.0147]

Table 2: **Table S2 | BNN hyperparameter search space and optimal values.**
表 S2 | BNN 超参数搜索空间与最优值。

Hyperparameter	Search range	Optimal (bnn-phy-mono)
Width	[64, 256]	254
Depth	[2, 8]	2
Learning rate	$[10^{-4}, 10^{-2}]$	9.81×10^{-4}
Weight decay	$[10^{-9}, 10^{-5}]$	8.09×10^{-8}
Batch size	{16, 32, 64}	32
Epochs	[200, 500]	399
Gradient clip	[0.1, 2.0]	0.674
w_{data}	[0.1, 5.0]	1.855
Prior σ	[0.1, 2.0]	0.478
KL weight	$[10^{-5}, 10^{-2}]$	1.93×10^{-4}
w_{mono}	[0.01, 1.0]	0.212
$\rho_{\text{abs,min}}$	[0.01, 0.5]	0.123
mono_topk	[10, 100]	53

Search: Optuna with 200 trials; objective: validation RMSE. Activation: SiLU throughout. Output heads: mean + log-variance (heteroscedastic).

Table 3: **Table S3 | BNN ablation study: four model variants on five primary outputs** (mean $\pm$ s.d., 5 seeds).
表 S3 | BNN 消融实验：四种模型变体在五个主要输出上的表现。

Model variant	Stress R^2	Stress PICP	k_{eff} R^2
Reference (ELBO only)	0.936 ± 0.007	0.995 ± 0.002	0.633 ± 0.260
Data-monotone	0.932 ± 0.005	0.998 ± 0.002	0.635 ± 0.260
Physics-regularized	0.935 ± 0.007	0.987 ± 0.007	0.633 ± 0.258
Data+inequality	0.924 ± 0.007	1.000 ± 0.001	0.617 ± 0.249

Table 4: **Table S4 | Complete Sobol indices (S_1 and S_T) for stress and k_{eff}** (physics-regularized BNN, $N_S = 4096$, 50 replications, 90% CI).
表 S4 | 应力和 k_{eff} 的完整 Sobol 指标。

Parameter	Stress S_1 [90% CI]	Stress S_T [90% CI]
E_{slope}	0.037 [−0.046, 0.100]	0.039 [0.038, 0.039]
$E_{\text{intercept}}$	0.548 [0.498, 0.572]	0.566 [0.562, 0.571]
ν	0.063 [0.034, 0.097]	0.068 [0.067, 0.069]
α_{base}	0.210 [0.171, 0.254]	0.214 [0.212, 0.217]
α_{slope}	0.038 [−0.013, 0.091]	0.047 [0.046, 0.049]
T_{ref}	0.019 [−0.025, 0.055]	0.032 [0.031, 0.033]
k_{ref}	0.065 [0.014, 0.118]	0.073 [0.072, 0.075]
SS316_α	−0.004 [−0.068, 0.034]	0.005 [0.005, 0.006]
Parameter	**k_{eff} S_1 [90% CI]**	**k_{eff} S_T [90% CI]**
E_{slope}	0.010 [−0.052, 0.062]	0.005 [0.004, 0.005]
$E_{\text{intercept}}$	0.008 [−0.065, 0.071]	0.006 [0.006, 0.007]
ν	0.029 [−0.035, 0.086]	0.018 [0.018, 0.019]
α_{base}	0.771 [0.747, 0.788]	0.767 [0.758, 0.776]
α_{slope}	0.195 [0.112, 0.250]	0.198 [0.196, 0.201]
T_{ref}	0.010 [−0.023, 0.054]	0.011 [0.011, 0.012]
k_{ref}	0.000 [−0.063, 0.048]	0.006 [0.006, 0.007]
SS316_α	0.003 [−0.059, 0.074]	0.009 [0.009, 0.009]

Table 5: **Table S5 | Posterior calibration: 18 benchmark case summary** (physics-regularized BNN).
表 S5 | 后验标定：18 个基准算例汇总。

Case	Category	True stress (MPa)	Accept rate	90%-CI coverage	Post-mean err.
0	low	110.6	0.619	3/4	8.3
1	low	118.7	0.612	4/4	24.6
2	low	110.7	0.620	2/4	9.0
3	low	103.7	0.618	4/4	8.8
4	low	114.5	0.621	4/4	3.1
5	low	102.0	0.616	3/4	23.4
6	near	127.6	0.609	4/4	20.3
7	near	127.9	0.586	4/4	26.6
8	near	127.0	0.611	4/4	5.9
9	near	123.5	0.613	4/4	3.5
10	near	128.9	0.609	3/4	4.5
11	near	129.3	0.610	4/4	3.8
12	high	169.5	0.592	4/4	0.2
13	high	176.0	0.591	4/4	12.1
14	high	193.4	0.598	3/4	13.7
15	high	163.4	0.591	4/4	3.6
16	high	142.8	0.582	4/4	10.5
17	high	197.6	0.583	4/4	5.4
Overall			0.58–0.62	66/72 = 0.917	MAE = 10.4

Table 6: **Table S6 | Computational cost breakdown.**
表 S6 | 计算成本分解。

Operation	Time	Platform
Coupled HF solve (1 case)	2266 s	EPYC 9654 / 96 cores
BNN single draw ($M = 50$)	1.29×10^{-2} s	GPU
BNN batched (batch=1024, per sample)	1.34×10^{-5} s	GPU
20,000-sample forward UQ	$\sim$0.3 s	GPU
Sobol (50 reps $\times$ 40,960 evals)	$\sim$minutes	GPU
18-case MCMC (8000 iter $\times$ 4 chains)	$\sim$minutes	GPU
Speedup (single draw vs. HF)	1.76×10^5	
Speedup (batched vs. HF)	1.69×10^8	

Table 7: **Table S7 | Physics consistency: monotonicity violation rates** (435 test samples, discrete $\pm\delta$ perturbation test).
表 S7 | 物理一致性：单调性违反率。

Physics pair (input $\rightarrow$ output, sign)	Ref. BNN	Data-mono	Phy-reg	Data+ineq
$E_{\text{intercept}} \rightarrow$ Max stress (+)	0%	0%	0%	0%
$\alpha_{\text{base}} \rightarrow$ Max stress (+)	0%	0%	0%	0%
$\alpha_{\text{base}} \rightarrow k_{\text{eff}}$ ($-$)	0%	0%	0%	0%
$k_{\text{ref}} \rightarrow$ Max stress ($-$)	0%	0%	0%	1.1%
SS316_$\alpha \rightarrow$ Fuel temp ($-$)	36.3%	39.5%	35.6%	53.6%

Inequality constraints (max $\geq$ average temp, stress ≥ 0): 0% violations for all four BNN variants.
不等式约束（最大 $\geq$ 平均温度，应力 ≥ 0）：四种 BNN 变体均为 0% 违反。

Table 8: **Table S8 | Epistemic vs. aleatoric uncertainty decomposition on test set.**
表 S8 | 测试集上的认知与偶然不确定性分解。

	Ref. BNN		Phy-reg BNN	
Output	Epi%	Ale%	Epi%	Ale%
k_{eff}	12.0	88.0	10.2	89.8
Max fuel temp	12.0	88.0	10.1	89.9
Max monolith temp	11.2	88.8	10.1	89.9
Max stress	27.7	72.3	29.6	70.4
Wall expansion	34.1	65.9	41.7	58.3

Table 9: **Table S9 | External baseline: full comparison of scoring rules.**
表 S9 | 外部基线：完整评分规则对比。

Model	Output	RMSE	R^2	CRPS	NLL	ECE
MC-Dropout	k_{eff}	0.0003	0.856	0.0002	-6.502	0.152
	Fuel temp	4.497	0.590	2.624	3.352	0.183
	Monolith T	5.061	0.590	2.951	3.472	0.182
	Max stress	7.659	0.934	4.519	3.592	0.118
	Wall exp.	0.0008	0.999	0.002	-3.831	0.197
Deep Ensemble	k_{eff}	0.0003	0.828	0.0002	-6.476	0.142
	Fuel temp	4.495	0.590	2.634	3.326	0.190
	Monolith T	5.061	0.590	2.962	3.439	0.185
	Max stress	7.644	0.934	4.502	3.586	0.112
	Wall exp.	0.0006	0.999	0.002	-3.841	0.197

Table 10: **Table S10 | Prior sensitivity** (6 configurations, physics-regularized BNN, case 06, near-threshold).
表 S10 | 先验敏感性（6 种配置，物理正则化 BNN，案例 06）。

Prior variant	Coverage (4 params)	Mean KL vs. canonical	Accept rate
Canonical	100%	0.00 (reference)	0.61
Diffuse ($2\times$)	100%	0.15–0.22	0.63
Tight ($0.5\times$)	50–83%	0.64–1.28	0.57
Flat (uniform)	100%	0.24–0.38	0.64
Shift ($+0.5\sigma$)	67–100%	0.06–0.36	0.60
Shift (-0.5σ)	83–100%	0.08–0.28	0.61

Table 11: **Table S11 | Noise sensitivity** (noise fraction 0.5%–10%, physics-regularized BNN, case 06).
表 S11 | 噪声敏感性（噪声分数 0.5%–10%）。

Noise fraction	$E_{\text{intercept}}$ post. std	Accept rate	Coverage
0.5%	1.26×10^{10}	0.58	100%
1.0%	1.28×10^{10}	0.59	100%
2.0% (default)	1.32×10^{10}	0.61	100%
5.0%	1.36×10^{10}	0.63	100%
10.0%	1.40×10^{10}	0.65	83%

Table 12: **Table S12 | Data efficiency** (2 seeds per fraction).
表 S12 | 数据效率（每个分数 2 个种子）。

Train fraction	N_{train}	RMSE (mean $\pm$ s.d.)	R^2 (mean $\pm$ s.d.)
25%	507	4.93 ± 0.04	0.768 ± 0.002
50%	1015	4.66 ± 0.11	0.782 ± 0.005
75%	1522	4.68 ± 0.01	0.784 ± 0.002
100%	2029	4.62 ± 0.08	0.783 ± 0.005

3 Supplementary Notes / 补充说明

3.1 Note A: BNN training details / BNN 训练细节

A.1. Evidence lower bound (ELBO) derivation / 证据下界推导

The BNN posterior over weights $\mathbf{w}$ is approximated by a mean-field variational distribution $q(\mathbf{w}) = \prod_i \mathcal{N}(w_i \mid \mu_i, \sigma_i^2)$. The variational parameters are optimised by minimising the negative evidence lower bound:

$$\mathcal{L}(\boldsymbol{\mu}, \boldsymbol{\sigma}) = \mathrm{KL}\big[q(\mathbf{w}) \,\|\, p(\mathbf{w})\big] - \mathbb{E}_{q(\mathbf{w})}\big[\log p(\mathcal{D} \mid \mathbf{w})\big] \tag{1}$$

where $p(\mathbf{w}) = \mathcal{N}(\mathbf{0}, \sigma_p^2 \mathbf{I})$ is the isotropic Gaussian prior. The KL divergence admits a closed form; the data-fit term is approximated by a single Monte Carlo sample per mini-batch (Bayes-by-Backprop[10]).

BNN 的权重后验由均场变分分布近似。变分参数通过最小化负证据下界来优化。重参数化技巧使梯度计算成为可能：$w = \mu + \sigma \cdot \epsilon$，其中 $\epsilon \sim \mathcal{N}(0, \mathbf{I})$。方差参数 $\sigma = \log(1 + \exp(\rho))$ 以确保正性，$\rho_{\text{abs,min}}$ 钳制 $\rho \geq \rho_{\text{abs,min}}$ 以防止后验坍缩。

A.2. Heteroscedastic observation noise / 异方差观测噪声

The network outputs both a mean $\mu(\boldsymbol{\theta})$ and a log-variance $\log \sigma_{\text{obs}}^2(\boldsymbol{\theta})$ per output channel:

$$\log p(y_j \mid \boldsymbol{\theta}, \mathbf{w}) = -\tfrac{1}{2}\big[\log \sigma_{\text{obs},j}^2 + (y_j - \mu_j)^2 / \sigma_{\text{obs},j}^2\big] \tag{2}$$

该异方差公式允许 BNN 建模输入依赖的观测噪声，将其与认知不确定性分离。

A.3. KL annealing schedule / KL 退火策略

The KL weight β is annealed from 0 to its target value kl_weight $= 1.93 \times 10^{-4}$ over the first 100 epochs: $\beta(t) = \min(t/100, 1) \cdot$ kl_weight.

KL 权重通过线性策略在前 100 轮退火，防止 KL 项在数据拟合项找到合理权重空间区域之前主导早期训练。

A.4. Physics-prior monotonicity constraints / 物理先验单调性约束

The physics-regularized BNN adds an auxiliary loss:

$$\mathcal{L}_{\text{mono}} = \frac{1}{K} \sum_{k=1}^{K} \max\big(0,\ -\text{sign} \cdot \partial\mu / \partial\theta_k\big) \tag{3}$$

weighted by $w_{\text{mono}} = 0.212$. Sign constraints are derived from constitutive physics (e.g. stress increases with $E_{\text{intercept}}$).

物理正则化 BNN 添加辅助损失项，惩罚违反从本构律推导的已知单调性关系。

A.5. Gradient clipping and initialisation / 梯度裁剪与初始化

Gradient norms are clipped at 0.674. Weight means μ are initialised with Xavier uniform; ρ values are initialised uniformly in $[-5, -4]$.

梯度范数按参数组裁剪至 0.674。权重均值以 Xavier 均匀初始化；ρ 值在 $[-5,-4]$ 上均匀初始化。

3.2 Note B: Sobol sensitivity methodology / Sobol 敏感性方法

B.1. Sobol variance decomposition / 方差分解

For a square-integrable function $f(\theta_1,\ldots,\theta_d)$ with independent inputs:

$$V=\sum_i V_i+\sum_{i<j} V_{ij}+\cdots+V_{1,\ldots,d} \tag{4}$$

The first-order Sobol index is $S_i=V_i/V$; the total-order index is $S_{T,i}=1-V_{\sim i}/V$.

一阶 Sobol 指标 $S_i=V_i/V$；全阶指标 $S_{T,i}=1-V_{\sim i}/V$。

B.2. Saltelli cross-matrix scheme / 交叉矩阵方案

Two independent $N\times d$ matrices $\mathbf{A}$ and $\mathbf{B}$ are drawn from the joint prior. For each input i, a cross-matrix $\mathbf{A}_{\mathbf{B}}^{(i)}$ is constructed. The model is evaluated at all $N(d+2)$ points.

从联合先验中抽取两个独立矩阵，对每个输入构造交叉矩阵，在所有 $N(d+2)$ 个点上评估模型。

B.3. Jansen estimator / Jansen 估计量

$$S_{1,i}=1-\frac{1}{2N}\sum_{j=1}^{N}\big[f(\mathbf{A}_{\mathbf{B}}^{(i)})_j-f(\mathbf{B})_j\big]^2/V \tag{5}$$

$$S_{T,i}=\frac{1}{2N}\sum_{j=1}^{N}\big[f(\mathbf{A})_j-f(\mathbf{A}_{\mathbf{B}}^{(i)})_j\big]^2/V \tag{6}$$

B.4. Replication-based confidence intervals / 基于重复的置信区间

The entire Sobol analysis is independently repeated $R=50$ times. The 90% CI is constructed from the 5th and 95th percentiles. The BNN posterior predictive mean ($M=30$ weight samples) serves as f.

完整 Sobol 分析独立重复 $R=50$ 次。BNN 后验预测均值（$M=30$）充当 f。

3.3 Note C: MCMC posterior calibration details / MCMC 后验标定细节

C.1. Likelihood formulation / 似然公式

$$\log p(\mathbf{y}_{\text{obs}}\mid\boldsymbol{\theta})=\sum_{j=1}^{J}\log\mathcal{N}\big(y_{\text{obs},j}\mid\mu_j(\boldsymbol{\theta}),\ \sigma_{\text{obs},j}^2+\sigma_{\text{BNN},j}^2(\boldsymbol{\theta})\big) \tag{7}$$

where μ_j is the BNN predictive mean ($M=20$ samples) and $\sigma_{\text{obs}}=2\%$ of the true value. $J=5$ primary outputs.

其中 μ_j 为 BNN 预测均值，σ_{obs} 为真实值的 2%。

C.2. Proposal distribution / 提议分布

Symmetric Gaussian random-walk: $\boldsymbol{\theta}^* = \boldsymbol{\theta}_{\text{current}} + \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\text{proposal}})$.

对称高斯随机游走提议，尺度通过 1000 次迭代的短试探运行校准。

C.3. Chain configuration / 链配置

Per benchmark case: 4 independent chains, $N_{\text{total}} = 8000$ iterations, burn-in 2000, thinning 5. Effective posterior: 1200 samples per chain, 4800 total. Calibrated parameters: $E_{\text{intercept}}$, α_{base}, α_{slope}, k_{ref}.

每个基准算例：4 条独立链，8000 次迭代，预热 2000 次，稀疏化步长 5。

C.4. Convergence diagnostics / 收敛诊断

Rank-normalised split-$\hat{R}$: $\hat{R}_{\text{max}} = 1.017$ (all 72 parameter–case combinations). $\text{ESS}_{\text{min}} = 352$. Acceptance rates: 0.58–0.62 (mean 0.605).

秩归一化 split-$\hat{R}$：$\hat{R}_{\text{max}} = 1.017$，$\text{ESS}_{\text{min}} = 352$，接受率 0.58–0.62。

C.5. Benchmark case selection / 基准算例选取

18 cases from the 435-sample test split: 6 low (102–119 MPa), 6 near (123–129 MPa), 6 high (143–198 MPa). Threshold: 131 MPa. Observations: true HF outputs $+2\%$ multiplicative Gaussian noise.

18 个基准算例取自测试集，覆盖低、阈值附近和高应力区间。

3.4 Note D: Forward UQ extensions / 前向 UQ 扩展

D.1. Predictive mean vs. full draw / 预测均值与完整抽样

The main-text forward UQ propagates through the BNN posterior predictive mean $y = \mathbb{E}_{\mathbf{w}}[\mu_{\mathbf{w}}(\boldsymbol{\theta})]$. As a consistency check, full predictive draws (sampling both weight uncertainty and observation noise) produce wider distributions but consistent means and percentile rankings.

正文前向 UQ 通过 BNN 后验预测均值传播。完整预测抽样分布更宽，但均值和百分位排名一致。

D.2. Latin hypercube sampling / 拉丁超立方采样

LHS ensures more uniform coverage of the input space than simple random sampling. $N = 20{,}000$ samples, $d = 8$ inputs.

LHS 确保比简单随机采样更均匀的输入空间覆盖。

3.5 Note E: Coupled simulation and material model / 耦合仿真与材料模型

E.1. Material constitutive models / 材料本构模型

SS316 monolith parameters grouped by physical function:

- Structural mechanics: $E(T) = E_{\text{slope}} \cdot T + E_{\text{intercept}}$; ν (Poisson's ratio). Stress: $\sigma = \mathbf{C}(E, \nu) \cdot \boldsymbol{\epsilon}$.

- Thermal deformation: $\alpha(T) = \alpha_{\text{base}} + \alpha_{\text{slope}} \cdot T$. Dual role: thermal strain $\epsilon_{\text{th}} = \alpha \cdot \Delta T$ and geometric expansion.
- Thermal transport: $k(T) = k_{\text{ref}} + k_{\text{slope}} \cdot (T - T_{\text{ref}})$.

SS316 一体化结构参数按物理功能分组：结构力学、热变形、热输运。

E.2. Picard iteration / Picard 迭代

OpenMC–FEniCS coupling: (1) initialise; (2) OpenMC computes k_{eff} and fission power; (3) FEniCS solves temperature, displacement, stress; (4) feedback updated geometry; (5) repeat until $\Delta T_{\text{max}} < 1\,\text{K}$. Under-relaxation factor: 0.1.

OpenMC–FEniCS 耦合迭代，收敛准则：$\Delta T_{\text{max}} < 1\,\text{K}$，欠松弛因子 0.1。

E.3. Output definitions / 输出定义

Each HF evaluation produces 15 scalar outputs: 7 decoupled (before coupling convergence) and 8 coupled (after full convergence, including k_{eff}). The posterior likelihood uses 5 primary outputs: k_{eff}, peak fuel temperature, peak monolith temperature, peak global stress, and wall expansion.

每次高保真评估产生 15 个标量输出：7 个解耦和 8 个耦合。后验似然使用 5 个主要输出。

E.4. Dataset generation / 数据集生成

$n = 2900$ samples by LHS; split (seed $= 2026$): training 2029, validation 436, test 435.

通过拉丁超立方采样生成 2900 个样本；固定划分复用以确保可重复性。

Supplementary References / 补充参考文献

9. Zhang T, Chen Y, Wang S, Liu X. Design-oriented multiphysics analysis and surrogate modelling of a heat-pipe-cooled reactor. *Energy*. 2025. (in press)
10. Blundell C, et al. Weight uncertainty in neural networks. ICML 2015; pp. 1613–1622.